2 Temperature, Pressure & Leak Troubleshooting

2.1 Overview

Helium loss on a Zero Boil-Off magnet should be insignificant (i.e., not measurable) over a period of weeks or months. If significant helium loss is suspected, the following is required to define and quantify the suspected loss:

- The magnet must be in a normal operating mode and stabilized after a ramp or fill.
- Two sets of data must be taken over time (2 days to 2 weeks, depending on the size of the leak) with known and calibrated instrumentation and at the same vessel pressure to establish the extent of the problem.

If high pressure or helium loss is established, use the troubleshooting guidelines that follow to determine the root cause(s) of the problem and to resolve the issue(s).

2.2 Preliminary Requirements

2.2.1 Tools and Test Equipment

Item	Part Number	Spare Part
Portable Liquid Helium Level Meter	2270571	Yes
Leybold Leak Detector	–	–
Portable RuO Temperature Monitor (included in Chapter 7, RuO Temperature Monitor Kit 2183710)	2171219	Yes
Portable Handheld Magnet Temperature Meter MTM-101	5119507	No
Plexiglas Heat Exchanger Cover Plate	2174214	Yes
Helium Gas Tubing	–	–
Coldhead Maintenance Kit	46-281088G3	Yes
Field Vacuum Pump Kit	46-294047G1	Yes
Magnet Monitor Pin Tester	2356202	Yes
M10 Allen Wrench	46-320470P10	Yes
Plus those items required by the procedure(s) needed to fix the problem.		

2.2.2 Consumables

Item	Part Number	Spare Part
Leak Detector Fluid, such as Bird Dog Quart Spray (P/N DOG-1Q), Big Blue (P/N Rt100S or Rt100G) or Snoop	–	–
Small Balloon	–	–
Small Sponge	–	–
Plus those items required by the procedure(s) needed to fix the problem.		

2.2.3 Replacement Parts

Item	Part Number	Spare Part
O-Rings included in Field Spare Parts Kit	5265652	–
Transition Flange O-Ring	46-281247P1	Yes
Plus those items required by the procedure(s) needed to fix the problem.		

2.2.4 Safety

No general safety information.

2.2.5 Required Conditions

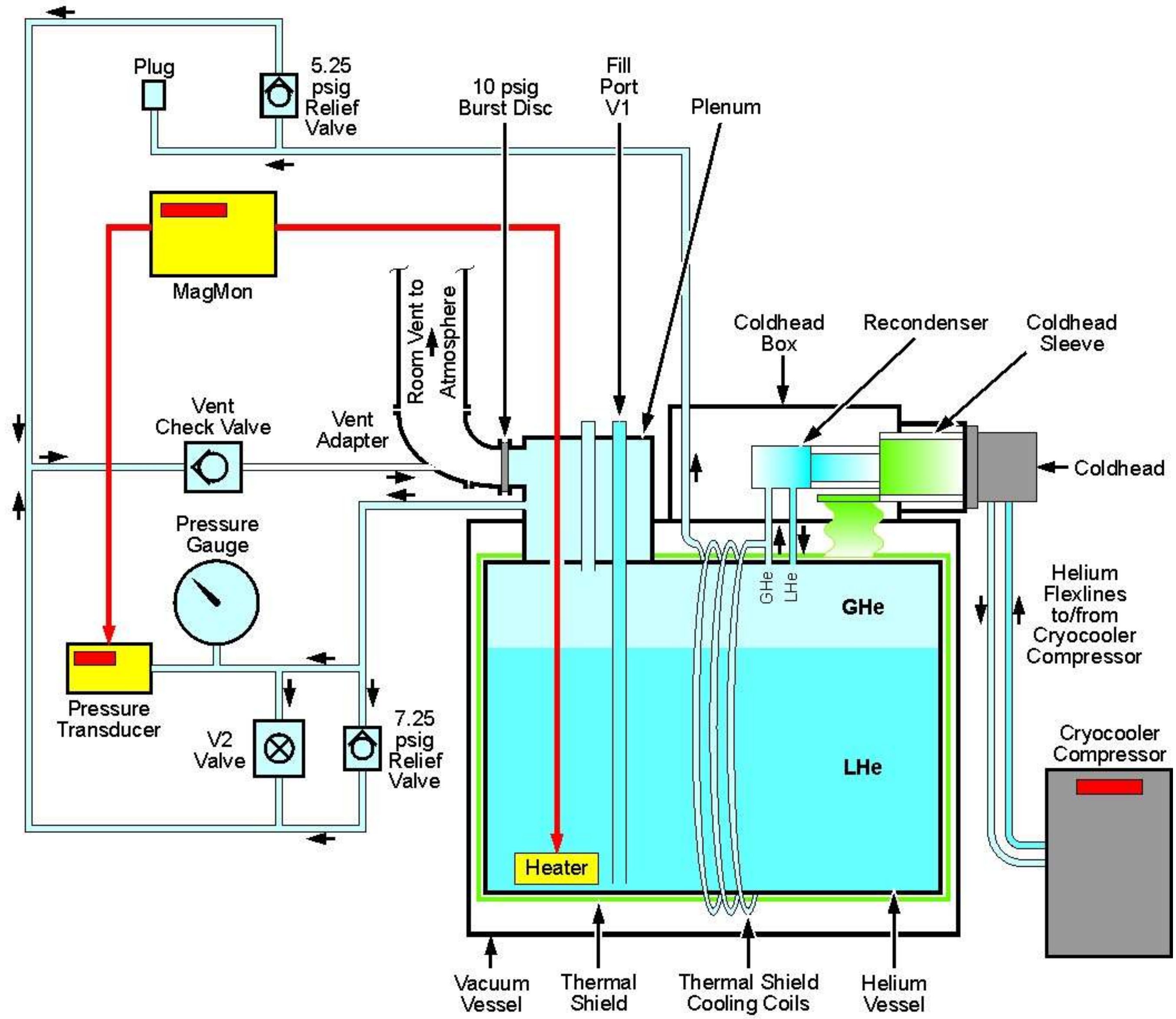
Condition	Reference
Removal of some magnet enclosure components may be required to complete the Temperature, Pressure and Leak Troubleshooting. If cover removal is required, refer to the appropriate: • Reference (to the right), and • Service Methods documentation (available through the Common Documentation Library at gehealthcare.com or your local GE Healthcare Service Representative)	Chapter 5, Upper and Side Wide-Open Enclosure Removal or Chapter 5, Introduction to HDe, HDx, MR450 and MR355/MR360 Enclosures

2.3 Procedure

2.3.1 Troubleshooting Guide

The illustration below shows a thermal system schematic of an installed magnet, including the path of liquid helium (LHe) and gaseous helium (GHe) through the helium vessel and plumbing.

Illustration 4-1: Magnet Thermal System Schematic



NOTICE

Follow the temperature and pressure troubleshooting procedures in sequence to minimize unnecessary Coldhead removal/replacement. Repair all problems as discovered, then proceed with additional troubleshooting as required.

Illustration 4-2: Initial Troubleshooting

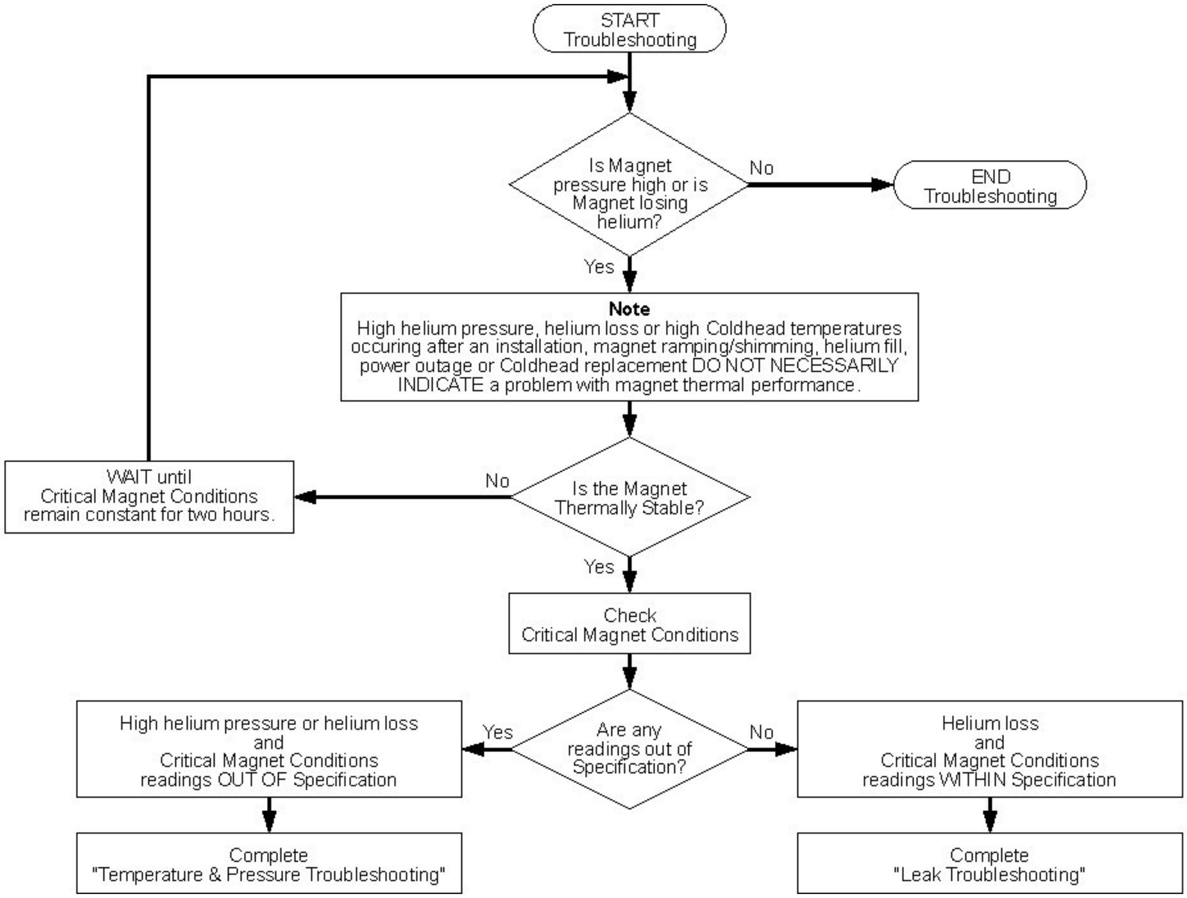


Table 4-2: Critical Magnet Conditions

Condition	Measure Used	Specification
Second Stage Coldhead Temp (RuO A)	MTM-101 or 2171219	<4.4K
First Stage Coldhead Temp (SI410 D1)	MTM-101 or 2171219	<45K
Recondenser Temp (RuO B)	MTM-101 or 2171219	>4.2K, but <4.5K
ΔT = RuO B - RuO A	MTM-101 or 2171219	>0.0K, but <0.2K
Magnet Helium Pressure	Magnet Monitor	4.0 psig ±0.2 psig

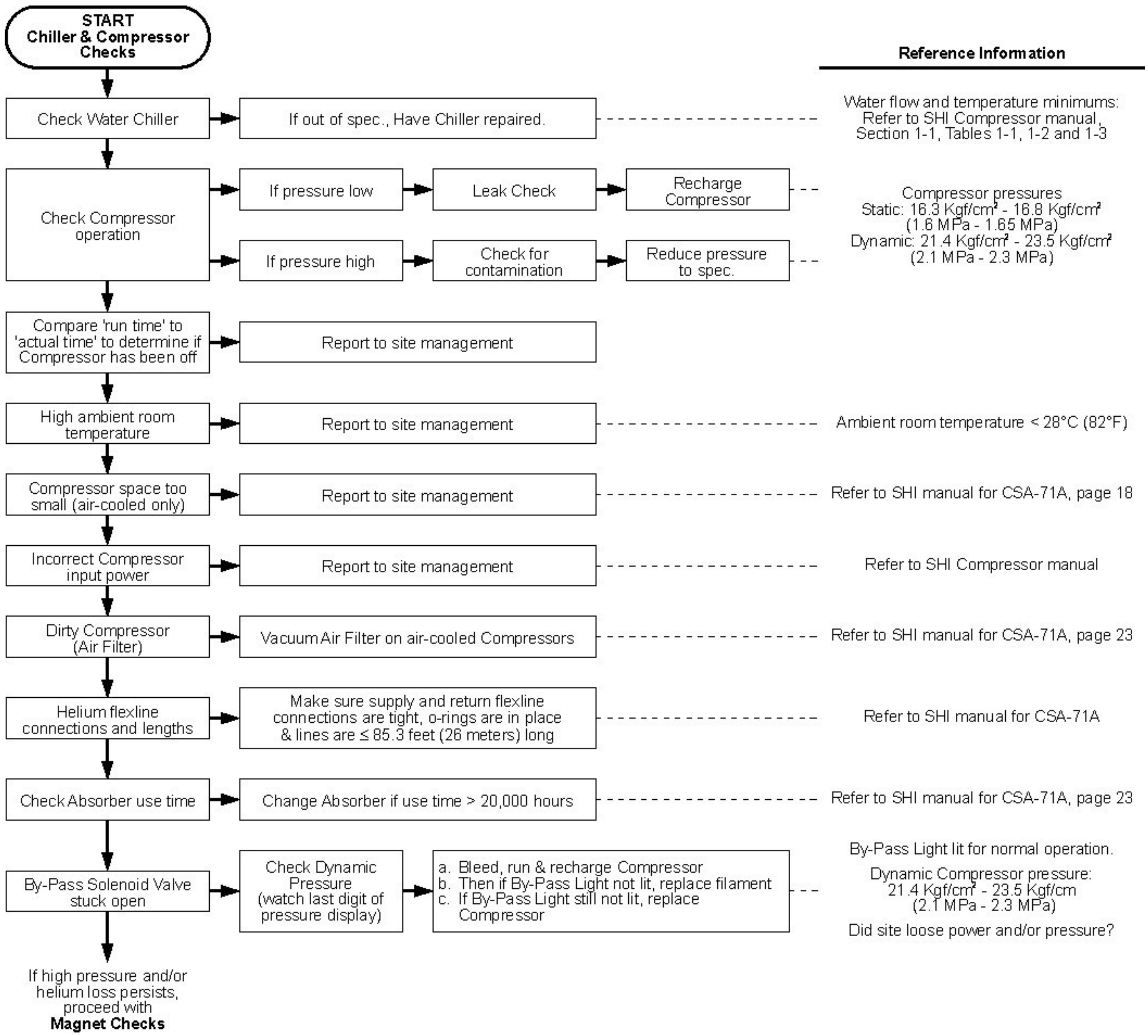
NOTICE

Second stage coldhead (RuO A) and Recondenser (RuO B) readings MUST be taken using a handheld RuO meter, either the MTM-101 (5119507) or Portable RuO Temp Monitor (2171219), connected directly to the RuO outputs on the Coldhead and Sleeve.

NOTE: Either temperature meter may show small oscillations while reading the second-stage coldhead resulting from displacer motion inside the Coldhead Sleeve and the monitor's sampling rate. Record the average temperature reading:
$(T_{min} + T_{max}) / 2$.

2.3.1.1 Chiller and Compressor Checks

Illustration 4-3: High Pressure/Temperature Troubleshooting Flowchart: Site Problems and Chiller/Compressor Checks



Manual	Vendor Part Number	GE Part Number
SHI SRDK Cryocooler Service Manual	CD32ZZ-055D	2262271
CSA-71A Compressor Unit Manual	CD32ZZ-060H	Included in 2262271

2.3.1.2 Magnet Checks

Use the flowchart in Illustration 4-4 in conjunction with the temperature-pressure graph in Illustration 4-5 to determine if vapor pressure is consistent with RuO readings.

Illustration 4-4: High Pressure/Temperature Troubleshooting Flowchart: Magnet and Coldhead Interface Checks (Part 1)

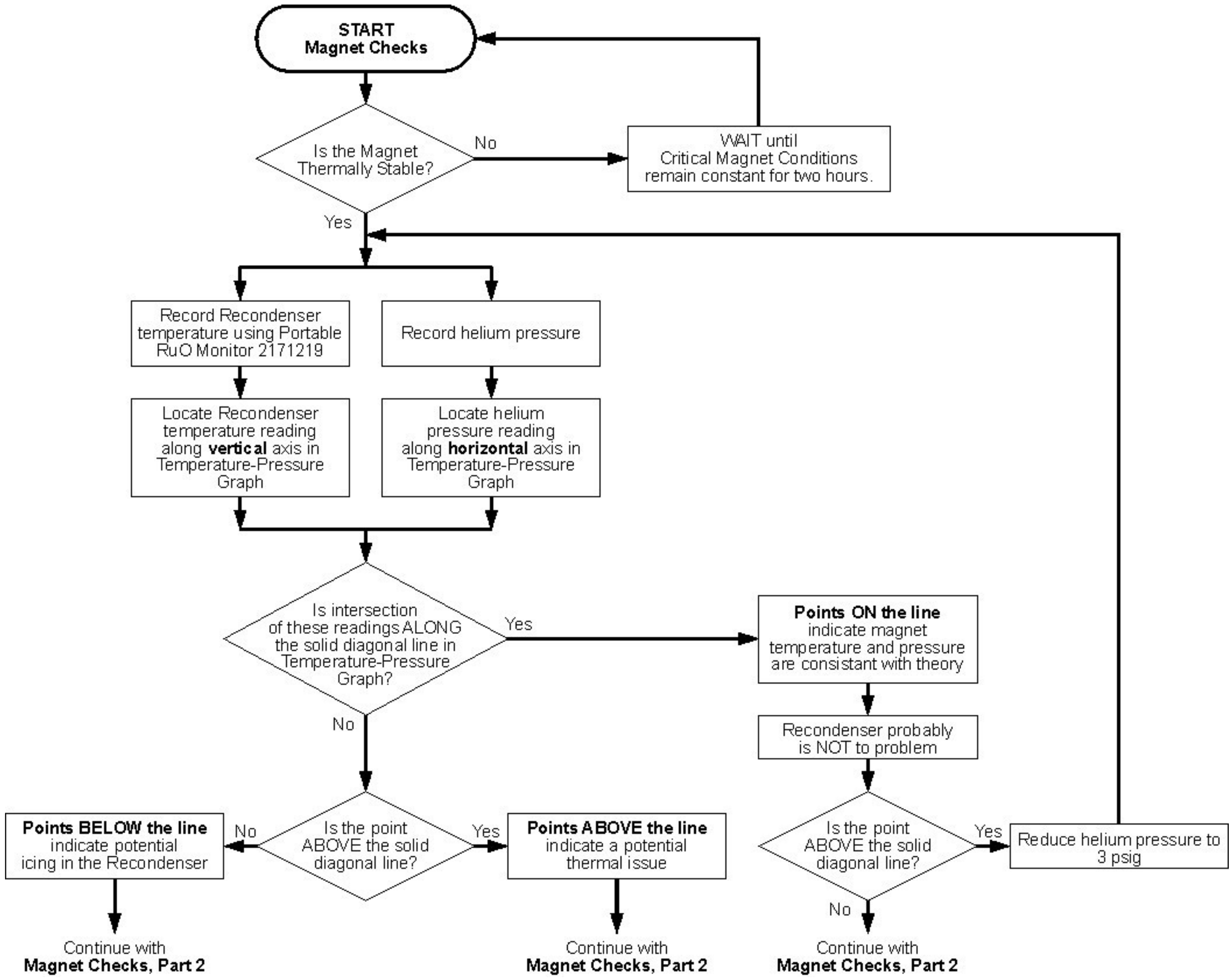
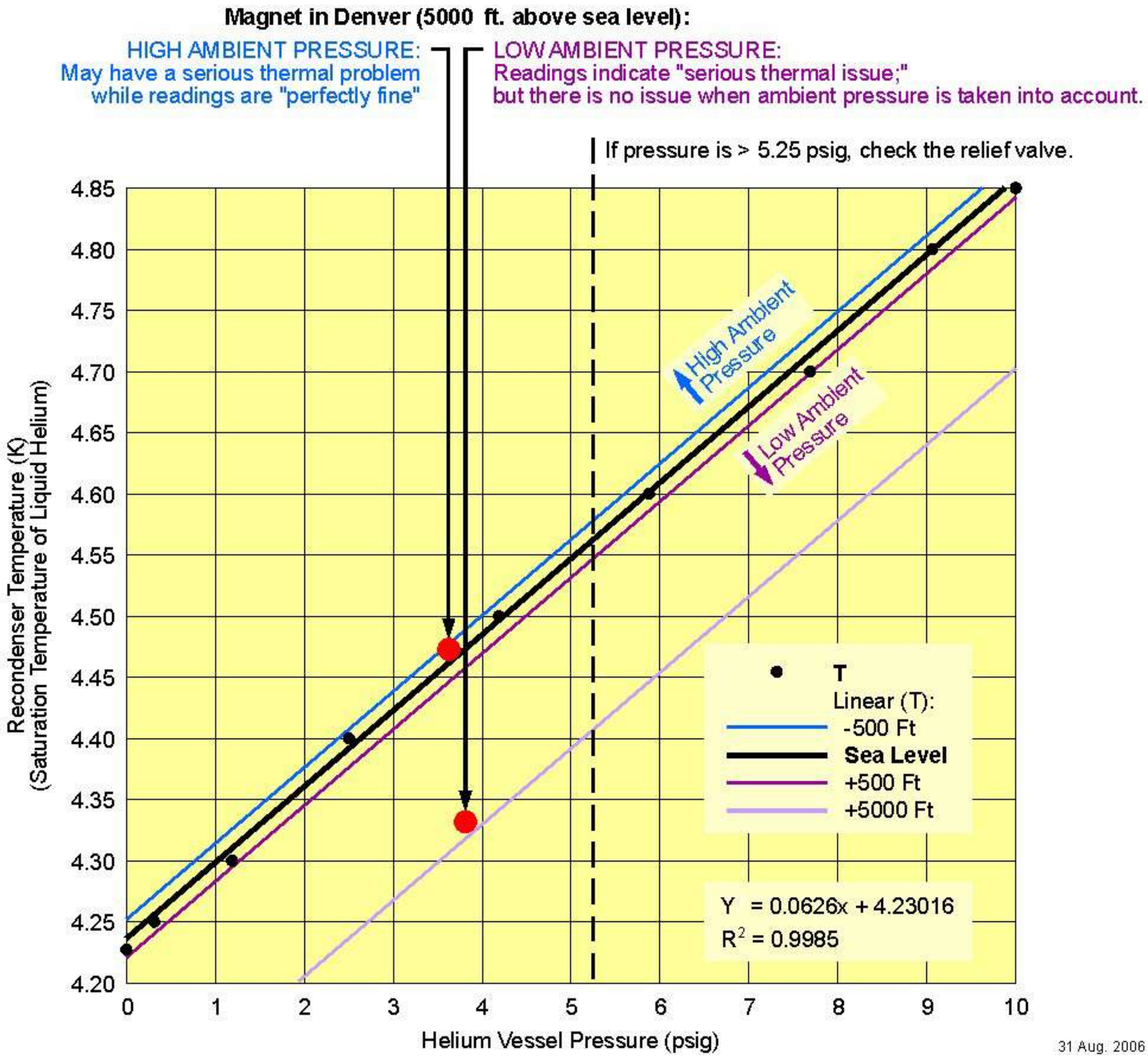


Illustration 4-5: Temperature-Pressure Graph: Liquid Helium (LHe) Saturation Temperature - Approximate Recondenser Temperature vs. Helium Vessel Pressure



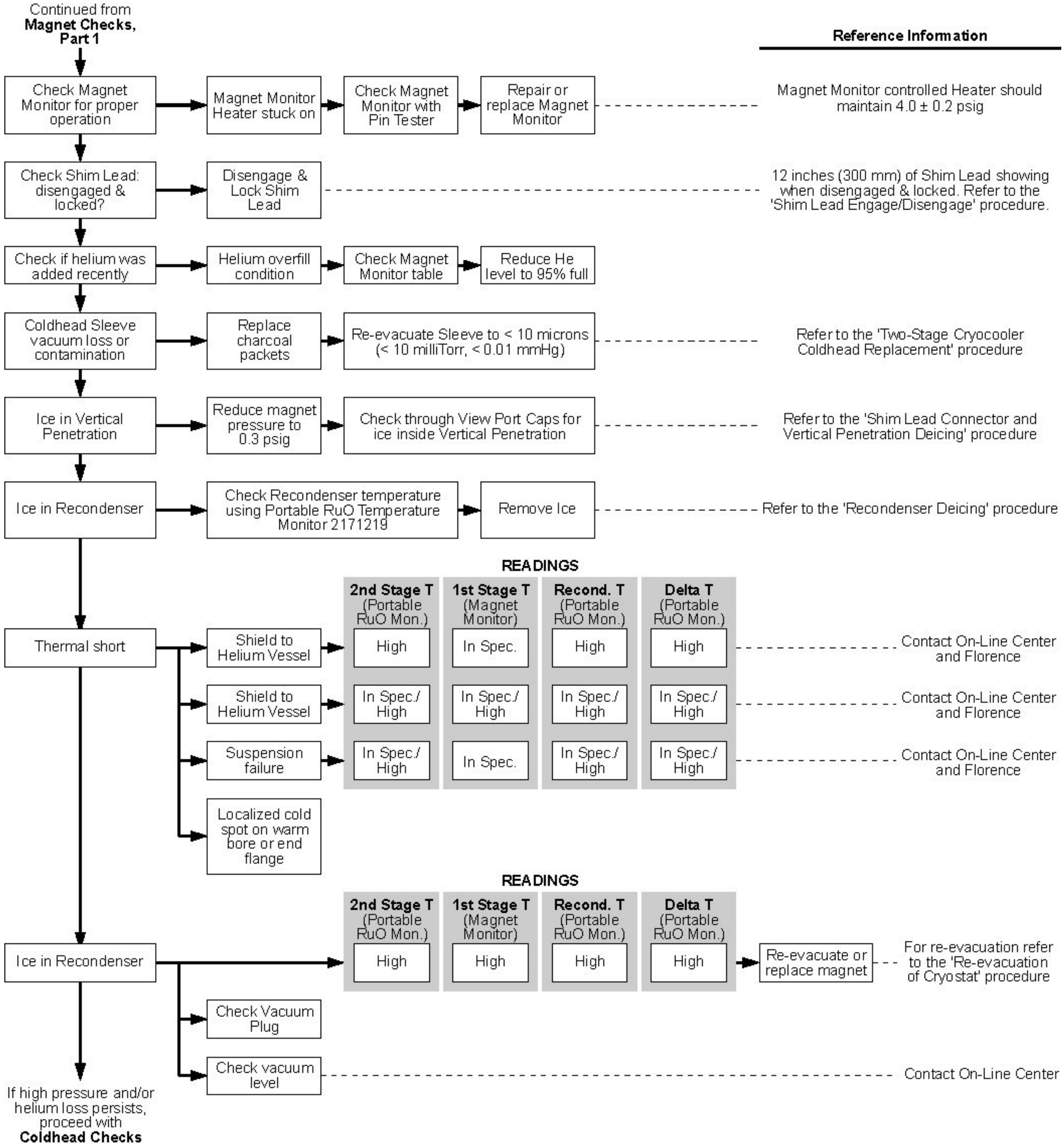
WARNING

POTENTIAL COLD BURN OR ASPHYXIATION HAZARD!
INTERNAL CRYOSTAT PRESSURE BUILD-UP TO >5.25 PSIG WILL OPEN THE 5.25 PSIG RELIEF VALVE.
BEFORE REMOVING EITHER RAMP OR FILL PORT CAPS OR LOOSENING ANY COMPONENT RESULTING IN THE RELEASE OF CRYOGEN PRESSURE:

1. **OBSERVE THE INTERNAL CRYOSTAT PRESSURE GAUGE READING,**
2. **SET MAGNET MONITOR PRESSURE TO 0.5 PSIG, AND**
3. **VENT CRYOSTAT DOWN TO 0.2 PSIG THROUGH VENT VALVE V2.**

NOTE: If the site has Magnet Monitor proprietary software, connect to the Magnet Monitor, and turn off the alarm for vessel pressure. Turn the vessel pressure control to a high value of 0.5 psig and a low value of 0.4 psig.

Illustration 4-6: High Pressure/Temperature Troubleshooting Flowchart: Magnet and Coldhead Interface Checks (Part 2)



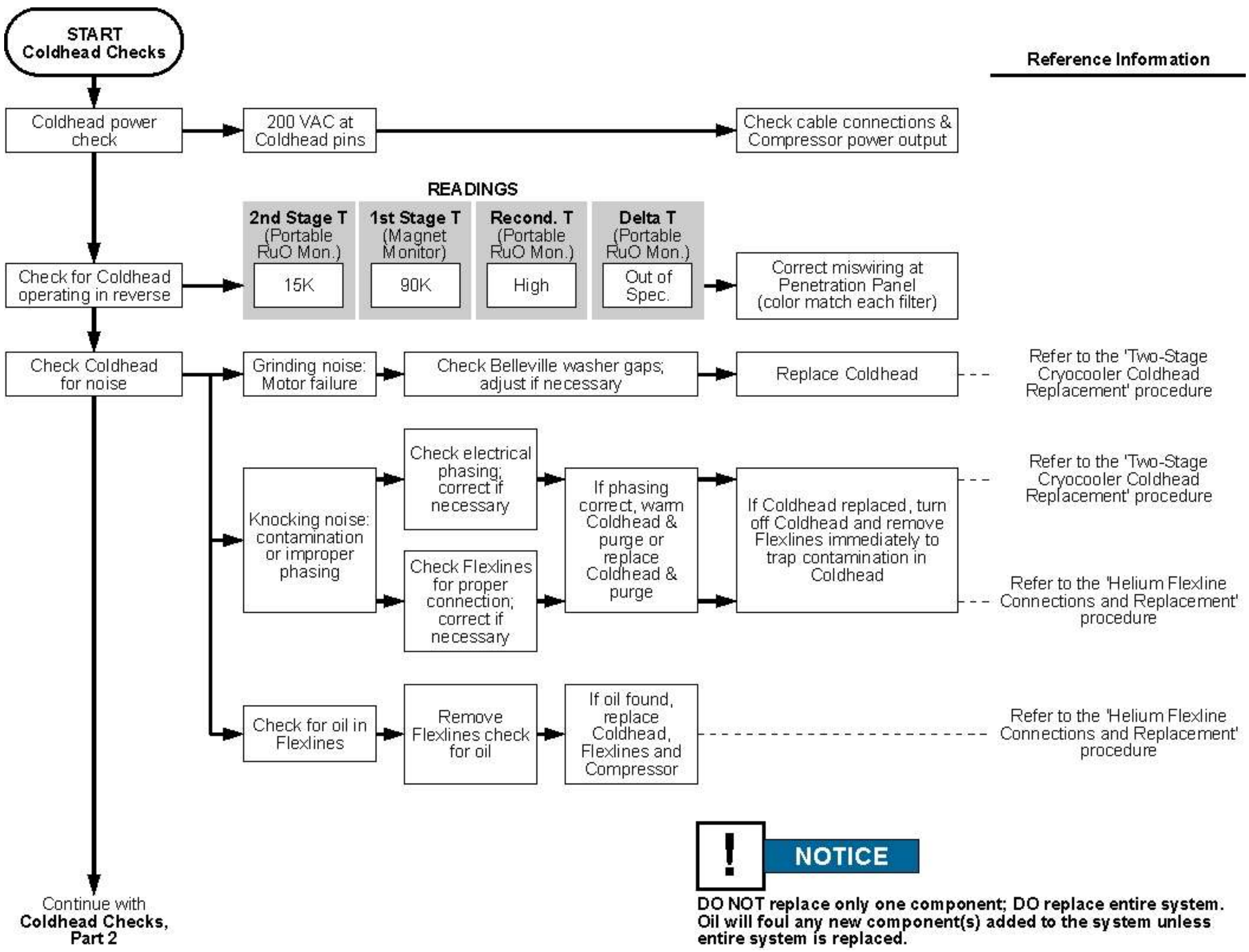
Procedures Referenced
Shim Lead Engagement and Disengagement
Chapter 5, Two-Stage Cryocooler Coldhead Replacement

Procedures Referenced
Shim Lead Connector and Vertical Penetration Deicing subsection in Chapter 5, Ice Removal
Recondenser Deicing subsection in Chapter 5, Ice Removal
Chapter 5, Re-Evacuation of Cryostat,

2.3.1.3 Coldhead Checks

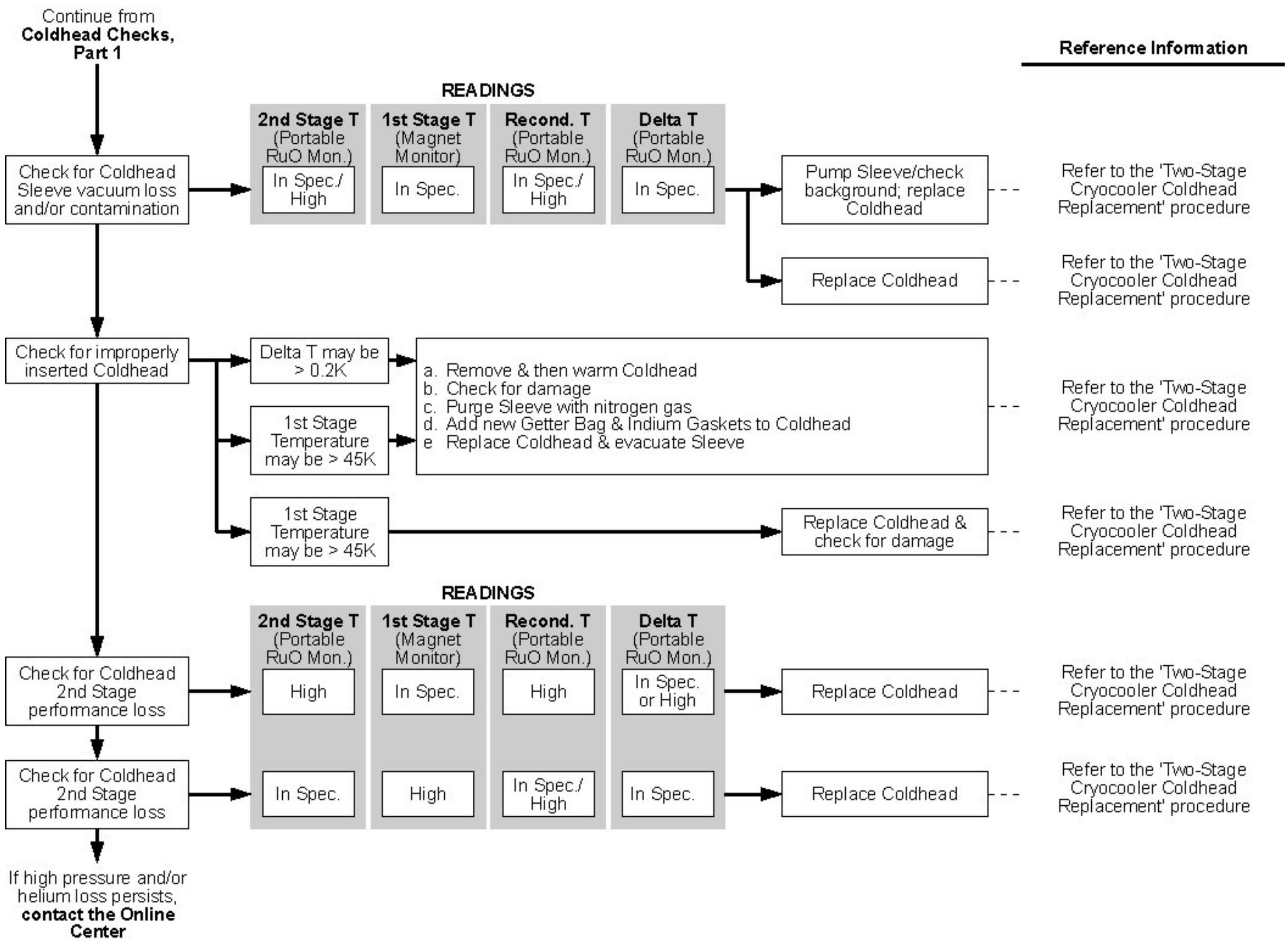
Only perform Coldhead Checks after completion of Chiller and Compressor Checks. Refer to the appropriate illustration below.

Illustration 4-7: High Pressure/Temperature Troubleshooting Flowchart: Coldhead Checks (Part 1)



Procedures Referenced
Chapter 5, Two-Stage Cryocooler Coldhead Replacement
Chapter 5, Helium Flexline Connections and Replacement

Illustration 4-8: High Pressure/Temperature Troubleshooting Flowchart: Coldhead Checks (Part 2)



Procedure Referenced
Chapter 5, Two-Stage Cryocooler Coldhead Replacement

2.3.2 Leak Troubleshooting

2.3.2.1 Introduction

Operation of the magnet in the C0BO (Controlled Zero Boil-Off) mode with internal helium pressure >4.1 psig increases the potential of helium leaks. A number of small helium leaks will have a significant effect on boil-off and result in the measurable loss of helium over time.

- Ice or condensation on plenum plumbing indicates a leak in the helium plumbing.
- Major helium leaks will depressurize the helium vessel and can result in cryopumping and icing inside the vessel.

Observe Magnet Monitor helium level data over a month or more. A slow helium level decrease is evidence of leaking helium.